SOMAIYA VIDYAVIHAR

K J Somaiya Institute of Technology

An Autonomous Institute Permanently Affiliated to the University of Mumbai

DEPARTMENT OF INFORMATION TECHNOLOGY

Name: Vaishnavi More

Roll Number: B - 72

Batch: SB4

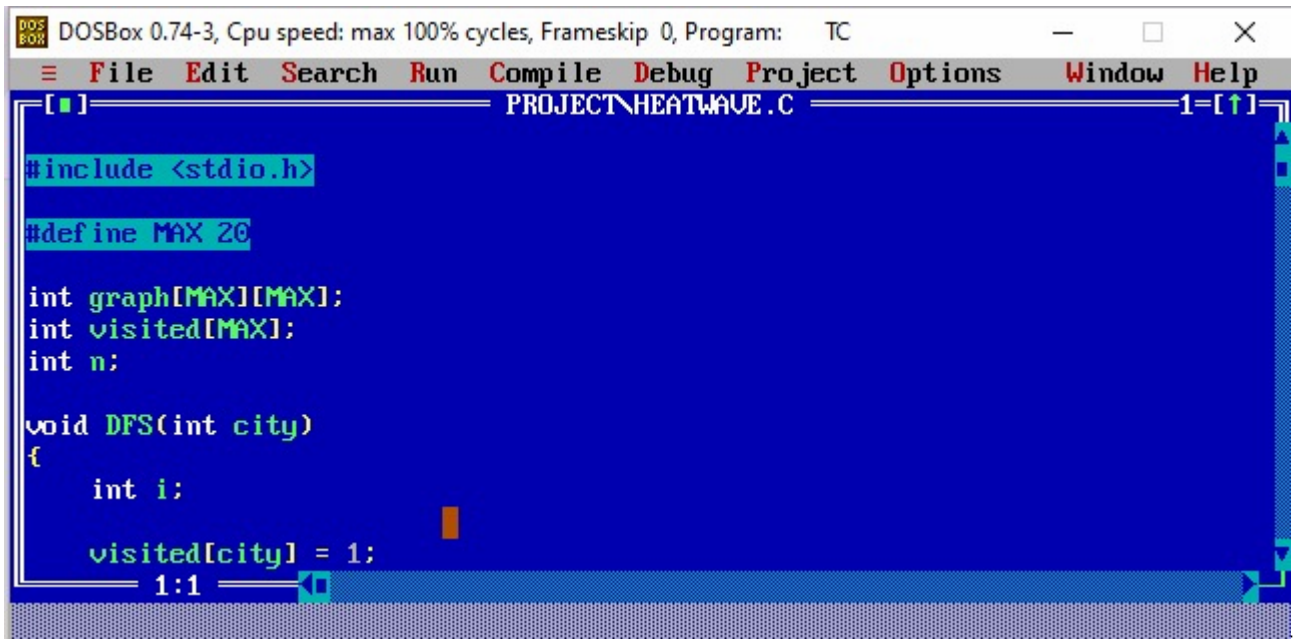
Course Name and Code: Data Structures Lab (ITL302)

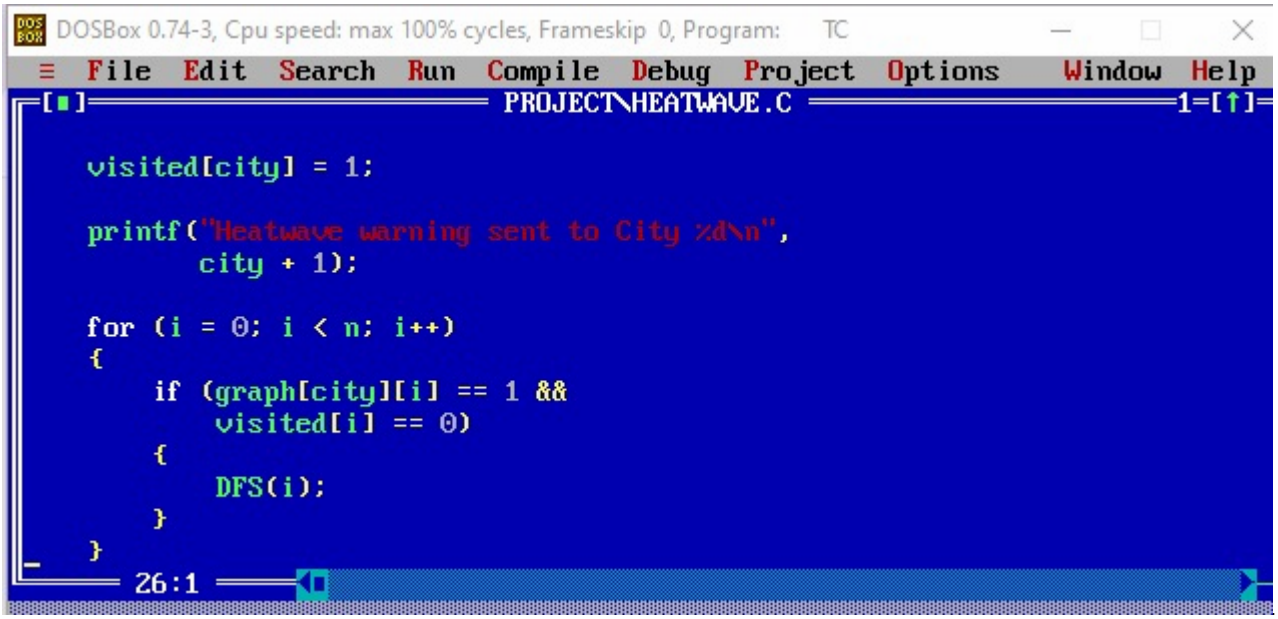
Semester: III (SYIT)

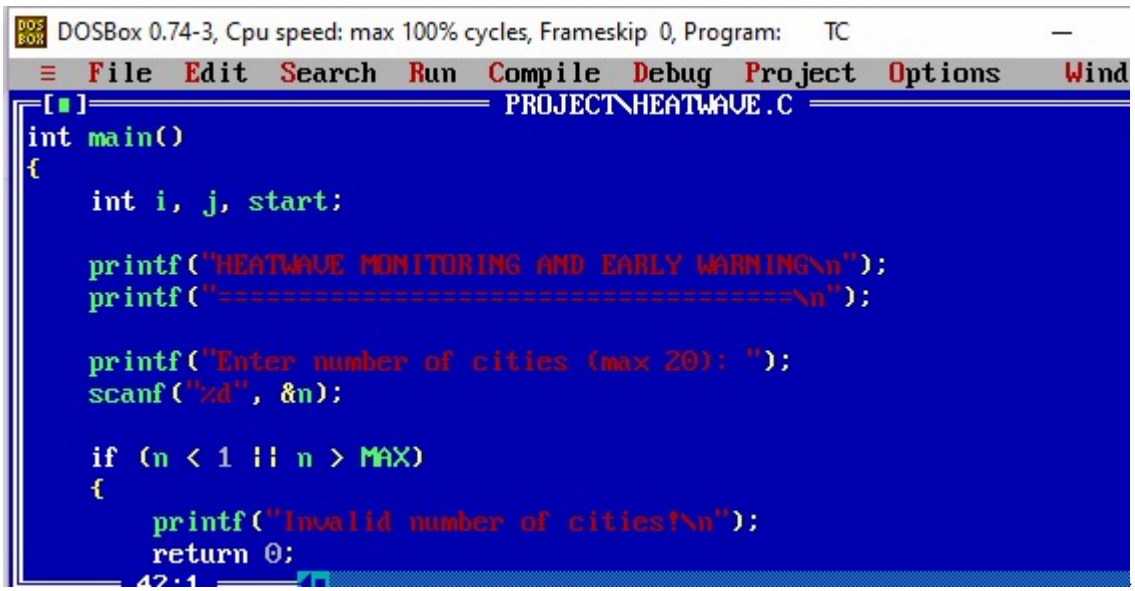
Academic Year: 2026-27 (Odd Semester)

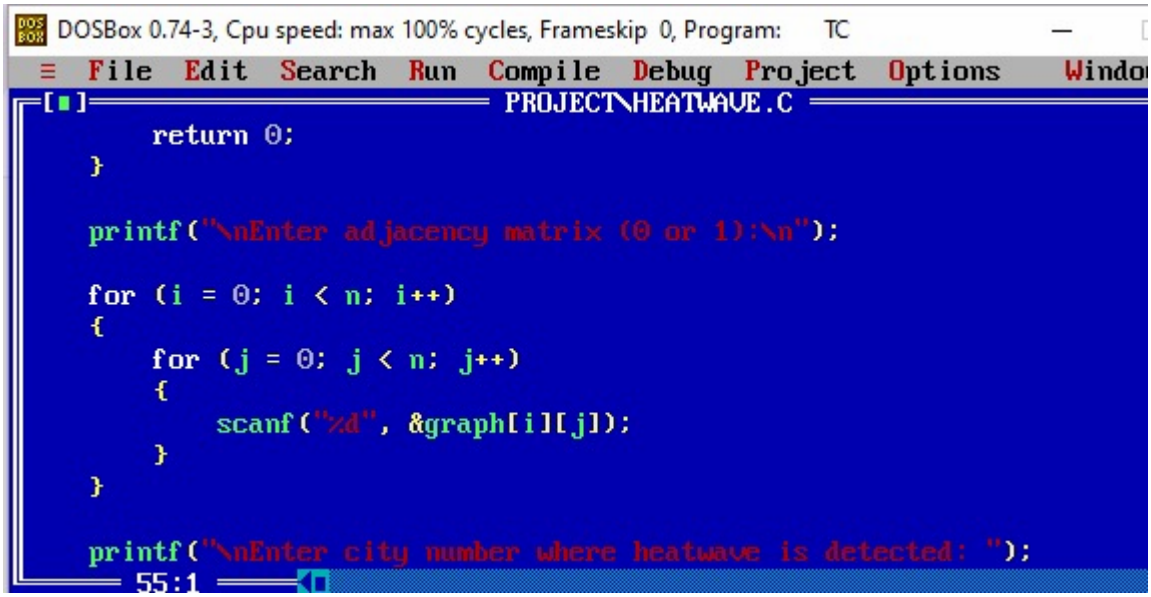
EXPERIMENT 7

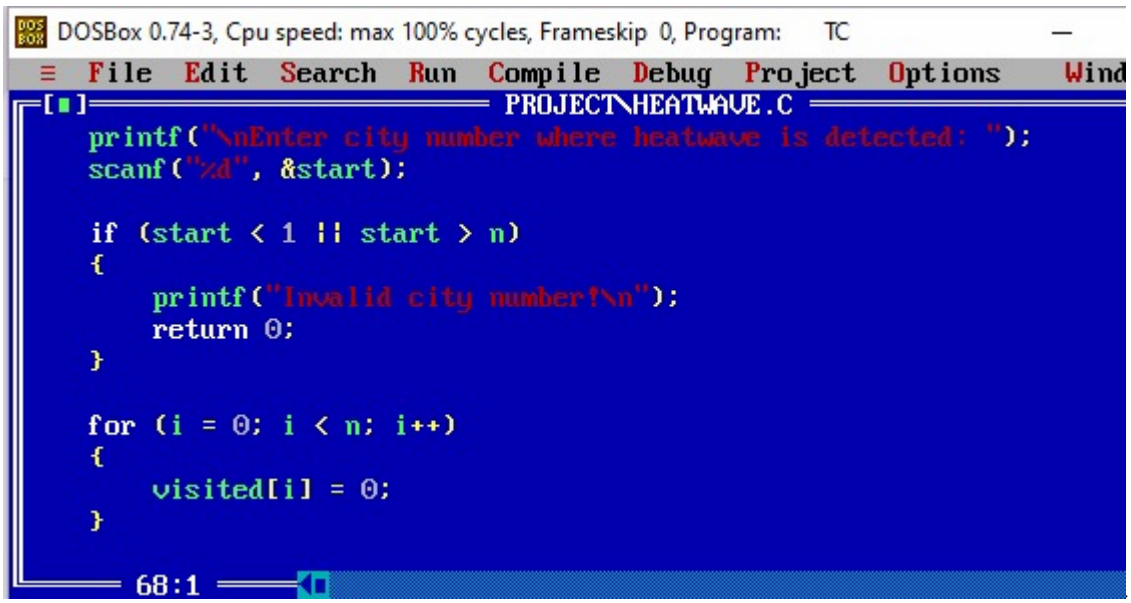
Code



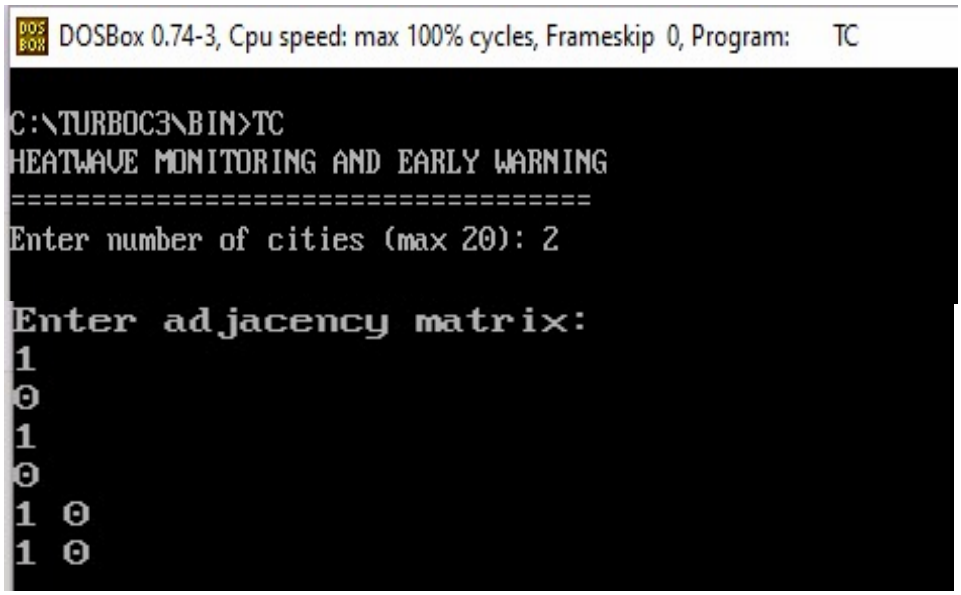

DOSBox 0.74-3, Cpu speed: max 100% cycles, Frameskip 0, Program: TC
File Edit Search Run Compile Debug Project Options Window Help
PROJECT\HEATWAVE.C
visited[city] = 1;
printf("Heatwave warning sent to City %d\n",
city + 1);
for (i = 0; i < n; i++)
{
if (graph[city][i] == 1 &&
visited[i] == 0)
{
DFS(i);
}
}
26:1

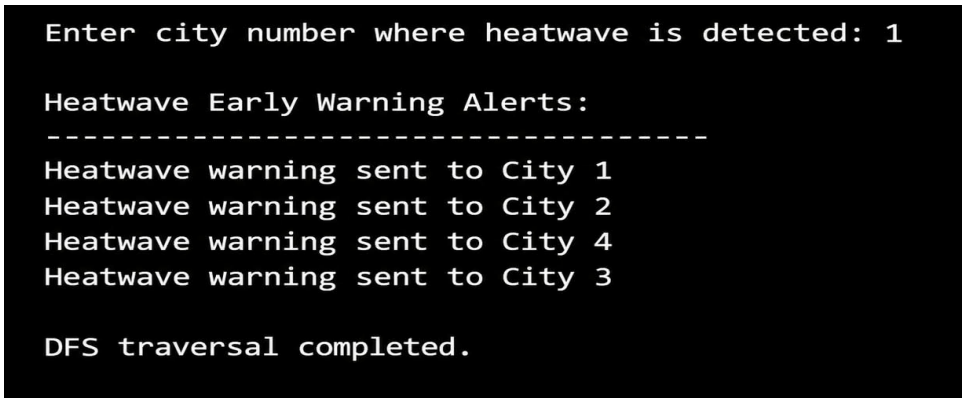

DOSBox 0.74-3, Cpu speed: max 100% cycles, Frameskip 0, Program: TC
File Edit Search Run Compile Debug Project Options Wind
PROJECT\HEATWAVE.C
int main()
{
int i, j, start;
printf("HEATWAVE MONITORING AND EARLY WARNING\n");
printf("=====================================\n");
printf("Enter number of cities (max 20): ");
scanf("%d", &n);
if (n < 1 || n > MAX)
{
printf("Invalid number of cities!\n");
return 0;
42:1


DOSBox 0.74-3, Cpu speed: max 100% cycles, Frameskip 0, Program: TC
File Edit Search Run Compile Debug Project Options Windo
PROJECT\HEATWAVE.C
return 0;
}
printf("\nEnter adjacency matrix (0 or 1):\n");
for (i = 0; i < n; i++)
{
for (j = 0; j < n; j++)
{
scanf("%d", &graph[i][j]);
}
}
printf("\nEnter city number where heatwave is detected: ");
55:1


DOSBox 0.74-3, Cpu speed: max 100% cycles, Frameskip 0, Program: TC
File Edit Search Run Compile Debug Project Options Wind
PROJECT\HEATWAVE.C
printf("\nEnter city number where heatwave is detected: ");
scanf("%d", &start);
if (start < 1 || start > n)
{
printf("Invalid city number!\n");
return 0;
}
for (i = 0; i < n; i++)
{
visited[i] = 0;
}
68:1

output


DOSBox 0.74-3, Cpu speed: max 100% cycles, Frameskip 0, Program: TC
C:\TURBOC3\BIN>TC
HEATWAVE MONITORING AND EARLY WARNING
==========
Enter number of cities (max 20): 2
Enter adjacency matrix:
1
0
1
0
1 0
1 0


Enter city number where heatwave is detected: 1
Heatwave Early Warning Alerts:
Heatwave warning sent to City 1
Heatwave warning sent to City 2
Heatwave warning sent to City 4
Heatwave warning sent to City 3
DFS traversal completed.